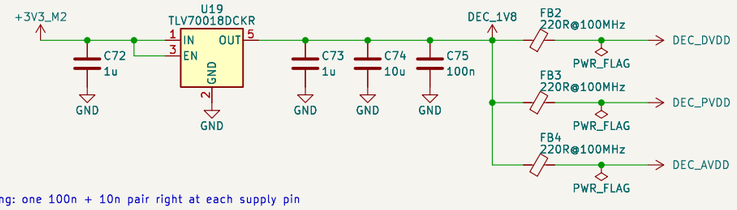
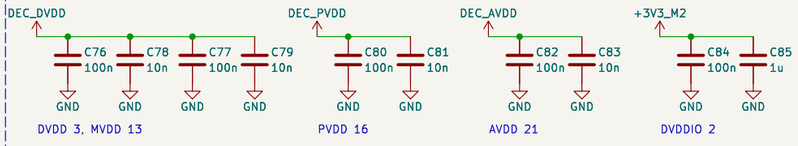
Decoder supplies



Decoupling: one 100n + 10n pair right at each supply pin



Notes

DRAFT — check against the ADI ADV7282-M datasheet before layout.
Circuit after Antmicro cvbs-mipi-bridge (Apache-2.0):
github.com/antmicro/cvbs-mipi-bridge

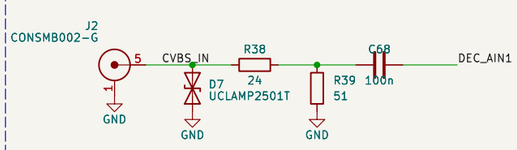
Raspberry Pi: dtoverlay=adv7282m (I2C 0x21 on the CSI/DSI bus, CAM1, 1 lane). Capture is V4L2 (/dev/video0), not libcamera. The overlay targets Unicam: CM4 only, untested on CM5.

I2C is mandatory: the chip starts with its CSI-2 output off and the driver configures it. Shares RPI_SDA/RPI_SCL with the MFi chip; no address clash (0x10 vs 0x21/0x42/0x44).

No CM GPIO used: PWRDWN and RESET come from RC delays. Supplies: 1.8 V LDO from +3V3_M2, one 220 R bead per 1.8 V rail.

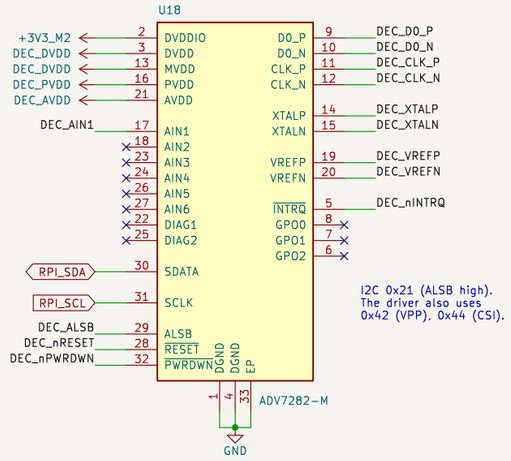
Camera power (+12 V from the reversing light) is not on this sheet. A short to +12 V on the video line would destroy D7: add series protection if the camera wiring is not trusted.

Camera input (CVBS)



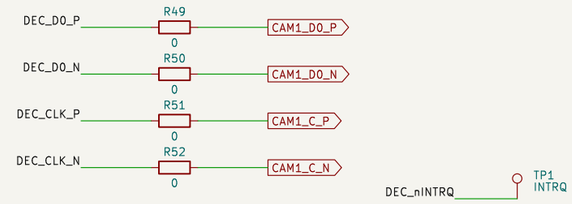
24 R + 51 R = 75 R termination; the divider scales the signal by 0.68 and the 100 nF AC-couples it into AIN1. Values from the Antmicro reference. Fit an RCA or Fakra jack instead if preferred. AIN2..AIN6 are free for more cameras: copy this network (select the input over I2C).

ADV7282-M video decoder



I2C 0x21 (ALSB high). The driver also uses 0x42 (VPP), 0x44 (CSI).

MIPI CSI-2 to CM CAM1 (1 lane)



CM pins: CAM1_D0_N/P = 115/117, CAM1_C_N/P = 127/129 (CN2 pads 15/17/27/29). Add the same four global labels on CN2 in the root sheet.

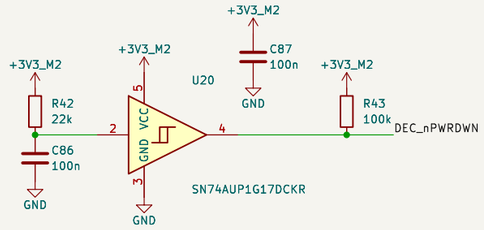
100 R differential pairs, P/N not swapped. The 0 R links let CAM1 take either this decoder or an FPC camera.

Address, I2C and reset sequencing



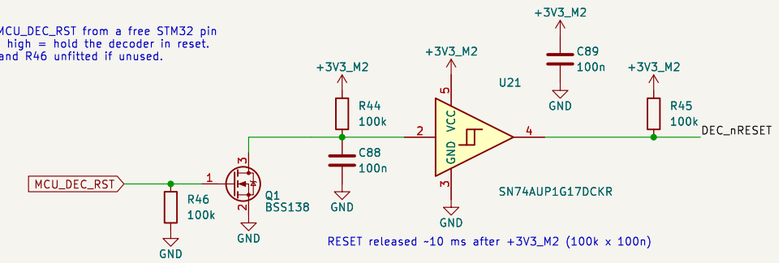
High = 0x21 (overlay default)

Optional pull-ups (DNP): the bus has none on this board.



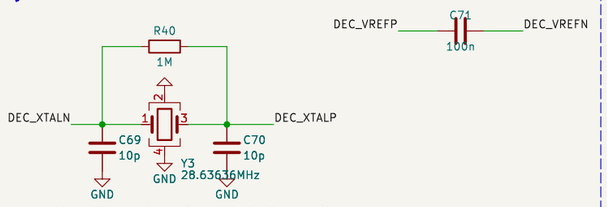
PWRDWN released ~2 ms after +3V3_M2 (22k x 100n)

Optional: MCU_DEC_RST from a free STM32 pin (e.g. PA4), high = hold the decoder in reset. Leave Q1 and R46 unfitted if unused.



RESET released ~10 ms after +3V3_M2 (100k x 100n)

Crystal and reference



Load caps from the Antmicro reference (18 pF crystal). C71 goes right at pins 19/20.

Not yet checked against the ADI datasheet
Circuit after Antmicro cvbs-mipi-bridge (Apache-2.0)
NavBridge CM4

Sheet: /		
File: Video_Decoder.kicad_sch		
Title: Video decoder — CVBS to MIPI CSI-2 (ADV7282-M)		
Size: A3	Date: 2026-11-01	**Rev: draft A**
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